SYSTEM REQUIREMENTS SPECIFICATION

Updated Functional & Non-Functional Requirements Document (Version 4)

PART I: REVISION LOG & SUMMARY OF CHANGES (v3 → v4)

1. What Was Added and Why

- **Clear Permission Rules:** The client told us exactly what each role can and cannot do. v4 adds the "cannot" rules, for example students cannot see other students' data, teachers cannot open other teachers' courses, and admins cannot change their own role.
- **Temporary Roles:** Admins can give a user a temporary role for a set period.
- **Faculty Assignment:** Admins can assign faculty to courses.
- **Course Analytics for Teachers:** Teachers can view analytics for their own courses.
- **Teachers Can Create Events:** Teachers (faculty) can now create campus events, together with Admins.
- **Pay Fees:** Students can pay their university fees online, and Admins handle those payments.
- **Degree Progress View:** Students see a progress bar for overall credits plus a checklist (curriculum map) of course requirements. The client chose this over a tree because a tree gets messy with 120+ credits.
- **Role-Based Dashboards:** Every role gets its own dashboard with real-time information, as the client described in their solution.
- **Single Cloud-Based Platform:** The client wants one cloud platform to replace spreadsheets, paper forms and disconnected software. This is now a stated requirement.
- **Future Enhancements & Open Items:** Two new parts list what the client wants in a later phase (Part IV) and what we still need to confirm (Part V).

2. What Was Modified and Why

- **Electricity Top-up:** The client dropped direct meter integration, so students pay from the wallet and an Admin enters the top-up by hand.
- **Waitlist:** Students are no longer enrolled automatically. When a seat opens, the student gets a notification and must confirm within 24 hours.
- **Timetable Conflicts:** Changed from "prevent" to a clear hard block. The system shows an error and disables the "Confirm Registration" button.
- **Campus Events:** Only Admins and Faculty can create events. Students can browse and sign up.
- **Grade Appeals:** Teachers submit grades, and Admins approve or reject appeals. v3 said "a teacher or admin".
- **Student Account Updates:** Students can update only non-sensitive personal information.
- **Mobile & Notifications:** The first release is a mobile-friendly website with in-app and email notifications. The native app with push notifications moves to a later phase.
- **Recommendations:** The first release uses clear rules (history, prerequisites, interests, degree requirements). AI-powered advising is planned for later.
- **Dashboards & Analytics:** Changed from one student dashboard to role-based dashboards for each user type.
- **Non-Functional Requirements:** Added the cloud-based platform requirement.

3. What Was Removed and Why

- **Direct Electricity Meter Integration:** Removed by the client because of high hardware cost and complexity and security risks. Replaced with manual Admin entry.
- **Student-Created Campus Events:** Removed to prevent spam and moderation problems.

- **Automatic Waitlist Enrollment:** Removed because students may no longer want the course or may have a clash. Replaced with a confirmation within 24 hours.
- **Mobile Push (moved to later):** Push notifications now come with the native mobile app in a later phase.
- **Attendance Tracking and Health & Support Services (still excluded):** Both stay out of scope for the same reasons as before.

PART II: FINAL REQUIREMENTS SPECIFICATION

BRANCH A: USER & AUTH MANAGEMENT

1. User Roles, Permissions & Authentication

- **Student:** Can register for courses, view their own grades, pay fees, view their own degree progress, use the wallet and update non-sensitive personal information. Cannot view other students' data, submit grades, or create courses or events.
- **Teacher (Faculty):** Can view rosters only for their assigned courses, submit grades, view analytics for their courses, upload course information and create campus events. Cannot change tuition fees, manage system-wide settings or access other teachers' courses.
- **Advisor:** Can approve or reject student course registrations, keep an eye on assigned students and give guidance. *(The client's answers did not cover this role, so its permissions need confirming. See Part V.)*
- **Administrator:** Can create courses, assign faculty, approve grade appeals, handle payments, manage user accounts, assign temporary roles and view all analytics. Cannot change their own role or bypass core academic rules such as prerequisite checks.
- **Authentication & MFA:** Users log in securely with Multi-Factor Authentication (required for Teachers, Advisors and Administrators, optional for Students) and can reset a forgotten password.
- **Session Management:** Sessions end automatically after a period of inactivity. Users can log out securely, accounts lock after repeated failed logins, and multiple logins on one account are controlled.
- **Role-Based Access Control (RBAC):** Each user can only use the functions that belong to their role.

2. Account Management & Admin User Management

- **Account Management:** Students can create an account and update their non-sensitive personal information, such as contact details. Sensitive details like Student ID, name and program are changed only through an Admin.
- **User Administration:** Admins can create, update, deactivate and reactivate student, teacher, advisor and administrator accounts.
- **Role & Advisor Assignment:** Admins can assign roles to users and assign an advisor to each student. Admins cannot change their own role.
- **Temporary Roles:** Admins can give a user a temporary role for a set period.

3. Registration Period Administration

- **Registration Period Management:** Admins can set, open and close the course registration window for each term.
- **Course Withdrawal Deadline:** Admins can set the last date for withdrawing from courses each term.
- **System Rule Management:** Admins can manage the global registration rules and limits, such as credit limits and waitlist limits.

4. Academic Advisor Requirements (Advisor Oversight)

- **Student Oversight:** Advisors can see the academic information and registration status of their assigned students.
- **Registration Review:** Advisors can approve or reject registration requests, and give a reason when they reject one.
- **Comments & Guidance:** Advisors can leave comments and suggest courses.
- **Advisor Overview:** Each advisor gets a simple list of pending requests and assigned students.

BRANCH B: ACADEMIC & REGISTRATION

5. Course Management Requirements

- **Course Creation:** Admins can add new courses.
- **Course Maintenance:** Admins can update a course's description, credits, schedule, capacity and rules.
- **Faculty Assignment:** Admins can assign faculty to courses.
- **Course Archiving:** Admins can archive courses that are no longer offered. A course can be permanently deleted only if it has no registration or grade history.
- **Capacity Management:** The system keeps track of seat limits and prerequisites for each course.

6. Teacher Requirements

- **Course Roster:** Teachers can see the class details and enrolled students for their assigned courses only.
- **Course Information:** Teachers can upload and update the course syllabus and other course information.
- **Grade Submission:** Teachers can submit grades and exam results for the courses they teach.
- **Course Analytics:** Teachers can view analytics for their own courses.
- **Campus Events:** Teachers can create campus events.

7. Course Information Requirements

The system provides the following course attributes:

- **Fields:** Course Name, Course Code, Credit Hours, Teacher, Prerequisites, Course Description, Available Seats, Class Schedule, Classroom Location.

8. Student Information Requirements

The system maintains the following student profile fields:

- **Fields:** Student ID, Name, Department, Program, Completed Courses, Current Courses, Academic Results, Credits Completed, Remaining Credits.

9. Course Registration Requirements

- **Search & Selection:** Students can search for courses by name, code, department, teacher or schedule, and pick the sections they want.
- **Prerequisite Checking:** The system checks prerequisites automatically and won't allow a registration if they aren't met. Nobody, including Admins, can bypass this check.
- **Conflict Prevention (hard block):** If two selected courses overlap in time, the system shows an error and disables the "Confirm Registration" button. The student must choose a different section or course. The same applies to registering for the same course twice.

- **Credit Limits:** The system enforces the minimum and maximum credit limits.
- **Registration Window:** Students can register only while the registration period set by the Admin is open.
- **Registration Confirmation:** After submitting, students see a confirmation and a summary of their registration.

10. Waitlist & Capacity Requirements

- **Seat Availability:** The system checks in real time whether seats are still available.
- **Waitlist Queue:** If a section is full, the student can join a waitlist, first come, first served.
- **Waitlist Offer (no automatic enrollment):** When a seat opens, the first eligible student is notified and must confirm within 24 hours. The student is not enrolled automatically.
- **Offer Expiry:** If the student does not confirm in time, the offer goes to the next student on the waitlist.
- **Waitlist Status:** Students can see their place in the queue and leave the waitlist whenever they like.

11. Course Drop & Withdrawal Requirements

- **Course Drop:** Students can drop a course any time before the Course Withdrawal Deadline.
- **Deadline Enforcement:** After the deadline, the system blocks the drop and tells the student why.
- **Seat Release:** When a student drops a course, the seat is freed and the waitlist process starts.

12. Timetable Requirements

- **Schedule Generation:** The system builds a timetable automatically from the courses a student has registered for.
- **Display Details:** Students can see their weekly classes with times and classroom locations.
- **Conflict Detection:** Overlapping classes are blocked at registration (see Section 9), and students are notified when their timetable changes.

13. Recommendation Requirements

- **Academic Analysis:** The system looks at a student's academic history and the courses they have already completed.
- **Interest & Rules Matching:** It also takes into account the student's interests, prerequisites and degree requirements.
- **Recommendation Display:** Students can see a personalised list of recommended courses.
- **Scope Note:** The first release uses these rules. AI-powered recommendations and advising are planned for a later phase (Part IV).

14. Exam, Academic History, CGPA & Grade Appeal Requirements

- **Exam Schedule:** Students can view published exam schedules.
- **Result Publication:** Teachers submit grades, and students can see them once they are published. Students can only see their own grades.
- **CGPA Calculation:** The system calculates GPA and CGPA automatically and keeps the academic history up to date.
- **Academic History:** Students can view their academic history and GPA/CGPA.
- **Degree Progress:** Students see a progress bar for overall credit completion and a checklist (curriculum map) showing which course requirements are done, in progress or still needed.

- **Grade Appeal / Correction Workflow:** Students can appeal a grade within a set appeal period. An Admin reviews the appeal and approves or rejects it with comments. An approved correction updates the result and CGPA, notifies the student and is recorded in an audit trail.
- **Appeal Status Tracking:** Students can follow the status of their appeal: submitted, under review, approved or rejected.

BRANCH C: CAMPUS SMART WALLET

15. Campus Smart Wallet & Payment Requirements

- **Pay for Food:** Students can pay at campus canteens and cafeterias by tapping their card or using the app.
- **Pay for Campus Bus:** Students can pay bus fares straight from their wallet.
- **Pay for Electricity:** Students can pay for their dorm room's electricity top-up from the wallet. There is no direct link to the electricity meter. After payment, an Admin enters the top-up into the system by hand.
- **Pay Fees:** Students can pay their university fees online. Admins handle and check these payments.
- **Wallet Balance Check:** Students can see how much money is left in their wallet, updated in real time.
- **Transaction History:** Students can see every payment and top-up, and filter them by date and type.
- **Recharge / Top-up:** Students can add money to the wallet through an online payment gateway and get a confirmation.
- **Insufficient Balance Handling:** If the balance is too low, the payment is declined and the student is told why.
- **Payment Confirmation:** Large payments need an extra check, such as a PIN or OTP.
- **Failed Transaction Reconciliation:** Failed or duplicate transactions are sorted out automatically, and refunds show up in the transaction history.
- **Audit Log:** Every wallet transaction is saved in a tamper-proof log that authorised administrators can check.

BRANCH D: CAMPUS LIFE & FACILITIES

16. Dormitory & Accommodation

- **Student Features:** Students can apply for a dormitory room and see which rooms are available and where their application stands.
- **Admin Features:** Admins can manage room allocations and accommodation status.

17. Library Requirements

- **Catalog & Borrowing:** Students can search for books, see what they have borrowed and keep track of due dates.
- **Reservations & Alerts:** Students can reserve books and get a notice when a book is overdue.

18. Campus Events

- **Events:** Admins and Faculty can create, update and cancel campus events. Students can browse and sign up for events but cannot create them.

19. Messaging & Announcements

- **Messaging:** Students, teachers and advisors can message each other directly.

- **Announcements:** Admins can post announcements to everyone or to a specific group.

BRANCH E: SYSTEM & OPERATIONS

20. Notification Requirements

- **Channels:** Notifications reach users through in-app alerts and email. Mobile push comes later with the native mobile app (Part IV).
- **Triggers:** Users are notified about registration deadlines, approvals and rejections, waitlist offers (with their 24-hour deadline), course changes, timetable changes, new recommendations, grade appeal status, wallet transactions and academic announcements.

21. Mobile Access & Push

- **Mobile Access:** Students can use registration and academic features from their phones through a mobile-friendly website.
- **Mobile Push (later phase):** A native mobile app with push notifications is planned for a later phase (Part IV).

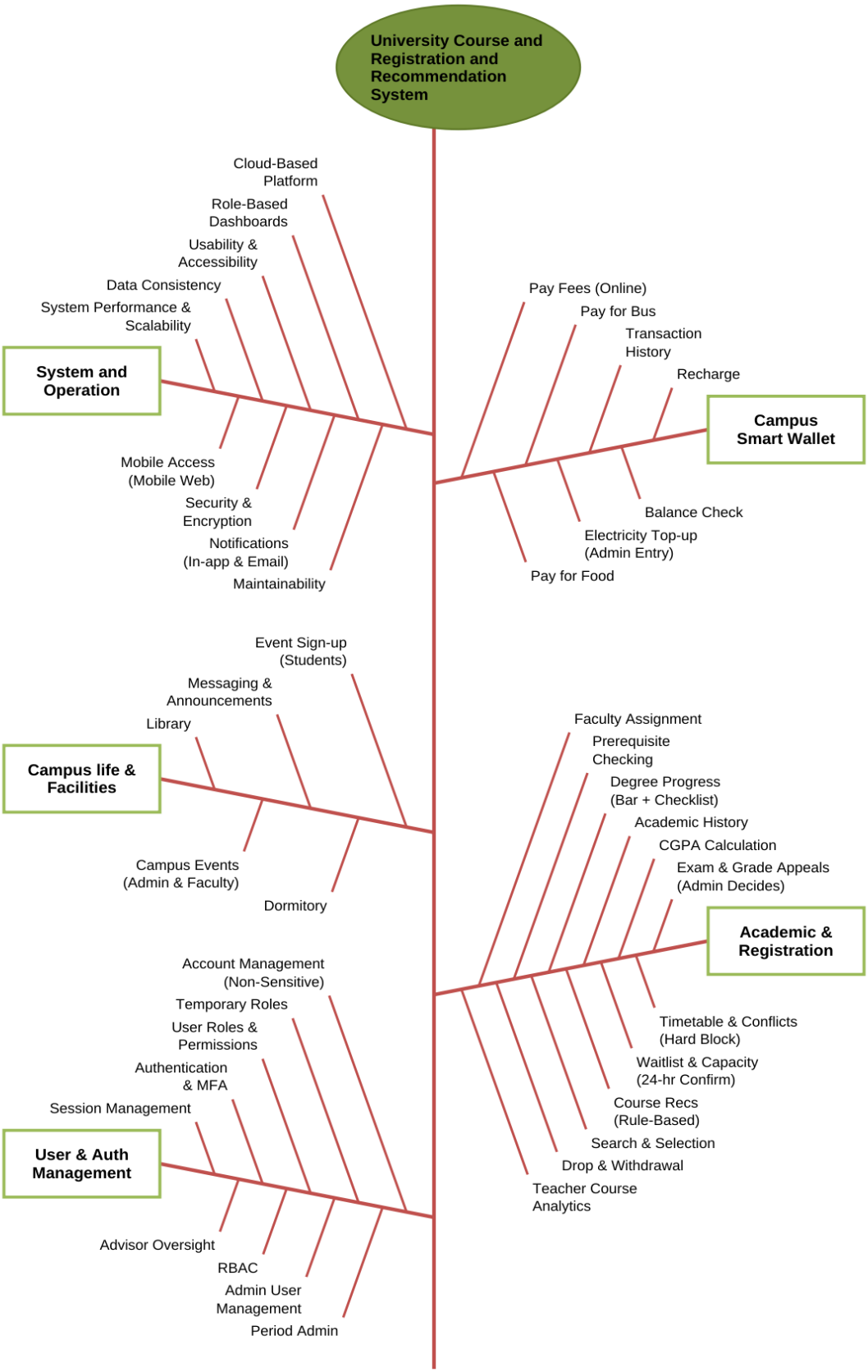
22. Analytics & Dashboard

- **Academic Insights:** The system looks at how a student is performing and shares useful insights.
- **Role-Based Dashboards:** Each role gets its own dashboard with real-time information. Students see registration status, timetable, results, degree progress and alerts. Teachers see their rosters and course analytics. Advisors see pending requests and their students. Admins see all analytics.
- **Admin & Advisor Reports:** Admins and Advisors can generate reports on registration trends, seat usage and student progress.

23. System Outputs

- **Outputs:** Registration Confirmations, Student Timetables, Course Recommendation Lists, Registration
- Summaries, Academic Summaries, Notifications and System Reports.

Feature Tree :


University Course and Registration and Recommendation System
System and Operation
Cloud-Based Platform
Role-Based Dashboards
Usability & Accessibility
Data Consistency
System Performance & Scalability
Mobile Access (Mobile Web)
Security & Encryption
Notifications (In-app & Email)
Maintainability
Campus Smart Wallet
Pay Fees (Online)
Pay for Bus
Transaction History
Recharge
Balance Check
Electricity Top-up (Admin Entry)
Pay for Food
Campus life & Facilities
Event Sign-up (Students)
Messaging & Announcements
Library
Campus Events (Admin & Faculty)
Dormitory
Academic & Registration
Faculty Assignment
Prerequisite Checking
Degree Progress (Bar + Checklist)
Academic History
CGPA Calculation
Exam & Grade Appeals (Admin Decides)
Timetable & Conflicts (Hard Block)
Waitlist & Capacity (24-hr Confirm)
Course Recs (Rule-Based)
Search & Selection
Drop & Withdrawal
Teacher Course Analytics
User & Auth Management
Account Management (Non-Sensitive)
Temporary Roles
User Roles & Permissions
Authentication & MFA
Session Management
Advisor Oversight
RBAC
Admin User Management
Period Admin

PART III: NON-FUNCTIONAL REQUIREMENTS

- **Platform:** One central cloud-based platform replaces the current mix of spreadsheets, paper forms and disconnected software.
- **Performance:** Everyday pages and searches should respond within 2 seconds, and a registration request should be processed within 3 seconds under normal load. The system must cope with the rush when registration opens, with a target of at least 5,000 users at the same time (to be confirmed against the real student intake).
- **Security & Encryption:** Data is encrypted while travelling (TLS) and while stored. Passwords are stored hashed and salted. Sessions are secure and access follows RBAC. Logins, grade changes, admin actions and financial transactions are all logged. Payments go through a compliant gateway, and full card details are never stored.
- **Reliability:** No data is lost during transactions. Backups run automatically at least once a day, and recovery targets are set by the university.
- **Data Consistency:** Registrations, seats, waitlists, grade changes and wallet payments are processed as complete, all-or-nothing steps. Even with many users at once, no seat is given out twice and no balance is spent twice.
- **Availability:** The system should be up 99.9% of the time while registration is open.
- **Usability:** The interface is easy to use on both desktop and mobile, so a new student can register without any training.
- **Scalability:** The infrastructure can grow with the student intake, without a redesign.
- **Maintainability:** The system is built in modules, so updates can go live without taking it offline.
- **Accessibility:** The system follows WCAG 2.1 Level AA so people with different abilities can use it.
- **Compatibility:** It works on current versions of the major browsers and on Android and iOS devices.